# Materials Science
# M.Sc. education program

crei.skoltech.ru/cest/education/materials-science-program

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# Where to get information

*Where to look:*

- crei.skoltech.ru/cest/education/materials-science-program

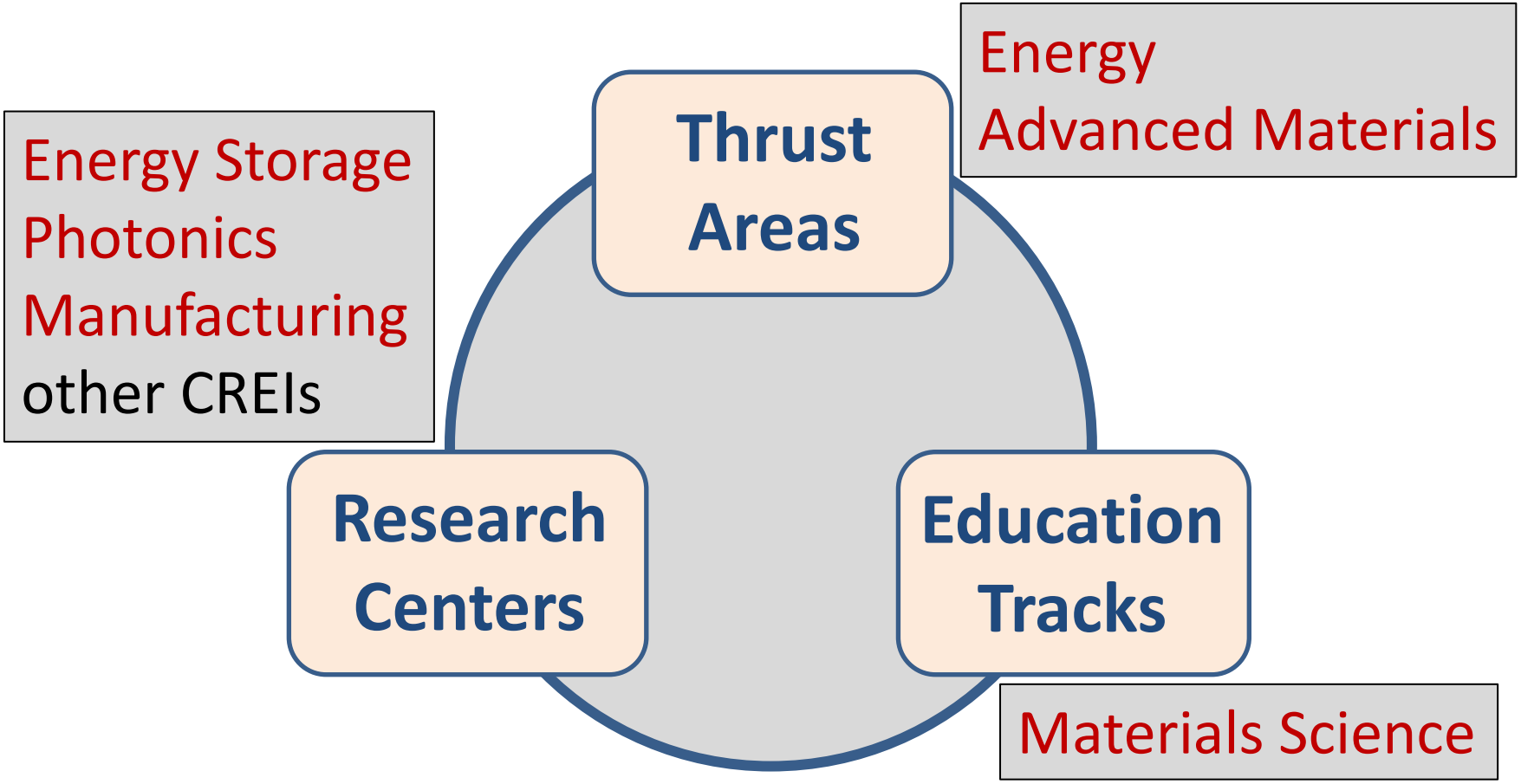
*Whom to ask:*

- Your academic/research advisor
- Program Coordinator – Andriy Zhugayevych
  - Experimental Materials Science – Artem Abakumov
  - Computational Materials Science – Andriy Zhugayevych
- Education Office

*For other contacts see the program web-page (above)*

# Skoltech research & education environment

- No separate theory or experiment departments
- No separate physics or chemistry departments



# Materials Science program: special courses

***Core courses***

- Survey of Materials **(Term 1B)**
- Materials Chemistry **(Term 2)**

***Experimental Materials Science***

***Recommended courses***

- Introduction to Solid State Physics**(2)**
- Material Structure Characterization Methods **(3)**
- Organic Materials for Electronics, Energy Conversion and Storage **(3)**
- Carbon Nanomaterials **(3)**
- Electrochemistry **(4)**
- Materials Selection and Design **(5)**

***Computational Materials Science***

***Core course***

- Computational Chemistry and Materials Modeling **(2)**

***Recommended courses***

- Structure & Property of Materials **(3)**
- Advanced Materials Modeling **(4)**
- Materials Selection and Design **(5)**
- Advanced Solid State Physics **(6)**

***Elective courses***

- discuss with your advisor

# Materials Science program: other activities

## Understand your load:

- 1 full course = 20 hours per week (except for Term 1B)
- *Hint*: combine research and education

## Seminars (1 hour per week):

- Research Seminar "Advanced Materials Science" – core
- Computational Materials Science seminar – for CMS track
- Your group seminars – per advisor
- Energy Colloquium – optional

## Background courses (half-course per term):

- Math for Engineers, Quantum Mechanics (Term 1B)
- Methodology, English, Writing, Presentation, Pedagogy

## Introductory courses from other programs (see webpage)

## Advanced courses (see webpage)

# M.Sc. Research in Materials Science

## In brief

- *Highly interdisciplinary*
- Experimental Materials Science with most publications on:
  - energy storage – Abakumov, Antipov, Fedotov, Nikitina, Stevenson
  - optoelectronics including solar cells – Nasibulin, Troshin
  - materials engineering – Korsunsky
- Computational Materials Science with most publications on:
  - methods – Gonze, Levchenko, Shapeev
  - applications – Oganov, Tretiak, Zhugayevych
  - theoretical chemistry and physics – Buchachenko, Fine
- *… with details on faculty web-pages: example*
- Research topics will be covered on Survey of Materials course

## M.Sc. Thesis 2016-2020

- Materials for energy conversion and storage (30 students)
- Materials for optoelectronics (5-10 students)
- Other materials and computational methods (10-15 students)

# Academic calendar and important deadlines

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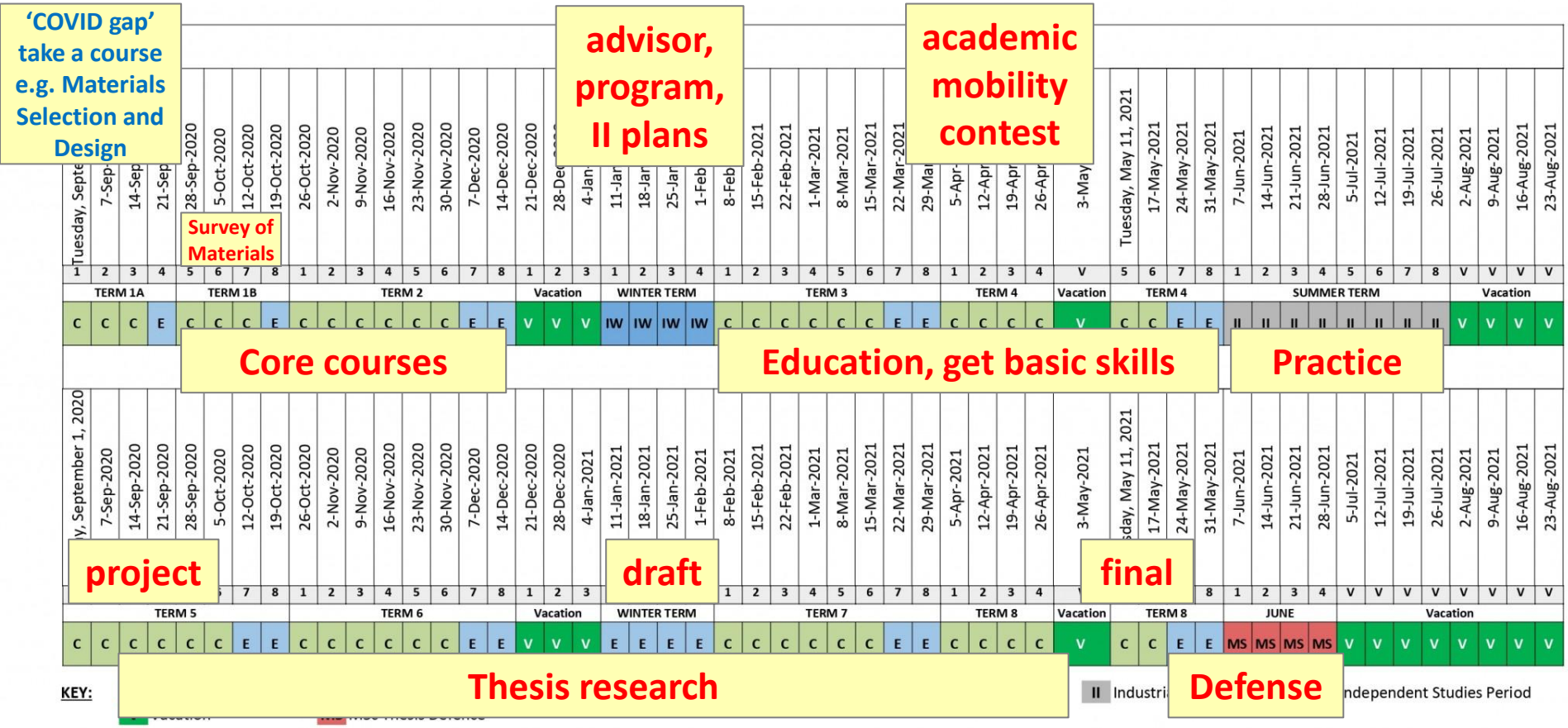
## Academic Calendar 2020-2021 Master's



# Approximate plan for the first 5 months

**Term 1A:**

- Materials Selection and Design course (20 hours per week)
- and/or 3/6-credit background course (if needed)

**Term 1B:**

- **Survey of Materials** course (40 hours per week)

**Term 2:**

- **Materials Chemistry** course (20 hours per week)
- Another 6-credit course (Intro to Solid State Physics, **Computational Chemistry and Materials Modeling** etc.)
- Optionally 3-credit research or elective course
- Seminars (at least 1 per week for 2 years)

**January:**

- Choose advisor and research field, plan next 6 months
- IW course (normally, ISP courses if needed or research)